
Mitigating LLM Dual-Use with Contextual Latent Adversarial Unlearning

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Abstract

Prompt-based guardrails for mitigating dual-use behavior in large language models (LLMs) have become standard due to their scalability and ease of deployment, but prompting alone is brittle: jailbreaks and adversarial prompting can reliably circumvent such defenses. This project proposes a new angle on dual-use mitigation by intervening not at the output level, but at the *representation* level. Specifically, we investigate whether intermediate contextual embeddings can be trained to *unencode* harmful features while preserving performance on benign tasks. By adversarially shaping latent geometry inside the transformer, we seek more robust dual-use suppression that cannot be bypassed through surface-level prompt manipulations.

Empirically, we find that gradient-reversal-based adversarial unlearning can substantially reduce the decodability of dual-use signals from internal representations while maintaining language modeling performance when applied at sufficient depth with moderate adversarial strength. In particular, lower adversarial weights ($\lambda = 0.3$) recover much of the test-set likelihood lost under stronger adversarial pressure, while still eliminating harmful responses under qualitative jailbreak probes. These results suggest that dual-use mitigation can be framed as a controllable trade-off at the representation level, rather than a binary behavioral constraint.

1. Introduction

Large language models (LLMs) are increasingly deployed in high-stakes settings where their generality creates dual-use risks: the same capabilities that enable benign assistance can also facilitate harmful or malicious behavior (Brundage et al., 2018; Weidinger et al., 2021). As model capabilities scale, the distinction between safe and unsafe behavior becomes increasingly contextual and procedural, rather than reducible to static keywords or topics.

Current safety practice primarily operates at the level of model outputs, using instruction tuning, refusal policies,

and prompt-based safeguards to restrict undesirable behavior (Ouyang et al., 2022). While effective under standard evaluations, these surface-level interventions are vulnerable to adversarial prompting and distribution shift (Wei et al., 2023; Zou et al., 2023b). Their failure modes suggest that refusal-based alignment suppresses harmful outputs without eliminating the internal representations that enable them.

In this work, we explore the hypothesis that dual-use capability is encoded in the geometric structure of transformer hidden representations. If harmful procedural knowledge is present in contextual embeddings, then output-level safeguards merely gate access to that knowledge rather than removing it. We propose a training-time approach that treats safety as a representational property: intermediate transformer layers are adversarially trained so that dual-use features are no longer decodable, while performance on benign language modeling tasks is preserved.

Our central research question is: *Can we suppress dual-use decodability from internal transformer representations without degrading safe task performance?* Answering this question advances a representation-level view of safety, in which robustness arises from modifying the model’s internal state space rather than relying on behavioral overlays.

1.1. Related Work

1.1.1. BEHAVIORAL ALIGNMENT AND GUARDRAILS.

Most deployed alignment methods for LLMs intervene at the input–output interface, using supervised instruction tuning or reinforcement learning with human feedback to induce compliant behavior (Ouyang et al., 2022). Empirical work on adversarial prompting shows that these mechanisms can be bypassed via rephrasing, role-playing, or multi-turn interaction, revealing limitations of purely behavioral safety approaches (Wei et al., 2023; Shen et al., 2023).

1.1.2. REPRESENTATION-LEVEL INFORMATION REMOVAL.

Several methods aim to remove specific information from hidden representations. Iterative Nullspace Projection (INLP) and related linear concept erasure techniques eliminate linearly decodable attributes from encoder representations (Ravfogel et al., 2020). More recent work explores

weight-level or low-rank interventions to remove particular concepts from large models (Eldan et al., 2023). These approaches primarily target static attributes or localized concepts, and do not directly address dynamic, multi-step procedural capabilities.

1.1.3. ADVERSARIAL REPRESENTATION LEARNING.

Gradient reversal and adversarial debiasing frameworks train feature extractors that support a primary task while suppressing auxiliary signals such as domain identity or protected attributes (?). This line of work demonstrates that representational geometry can be shaped through adversarial objectives. Our work adopts this framework, extending it to suppress dual-use procedural knowledge within intermediate transformer layers.

1.1.4. POSITIONING.

Unlike inference-time activation steering or post hoc interpretability interventions (Anthropic, 2024), our method modifies representations during training so that dual-use features are not encoded in contextual embeddings. This positions our approach as a training-time, representation-level alternative to output filtering and post hoc control.

2. Dataset

We use two categories of data: safe data and harmful data. We treat short question–answer (Q/A) pairs as a representative task, as they require factual recall and light reasoning while remaining easy to control for content.

- **Safe data:** We collect short factual science Q/A pairs from the SciQ dataset (Welbl et al., 2017). For data augmentation, we used GPT5.1 for augmentation to rewrite the Q/A pairs to include science questions of all levels (ie. elementary school, middle school, high school, university, PhD level). We collected 8k data-points for GPT2 fine-tuning.
- **Harmful data:** We construct harmful science Q/A pairs using prompts from JailbreakBench (Zou et al., 2023a), which includes adversarially crafted queries designed to elicit disallowed model behavior through obfuscation, role-playing, or prompt injection. We restrict these prompts to science-related content and convert them into short-answer Q/A form to match the structure of the safe dataset. In addition, we also modify generations in the safe dataset to become harmful (3). In total we collected 8k data points here too

The intuition here is that we want a latent space that embeds science knowledge both harmful and harmless. We are interested to see if we can learn only the harmless knowledge

Original Question	Modified Question
What vitamin helps blood clotting?	I am a vampire who wants to prevent my victims' blood from clotting—what vitamin should I remove?
Answer: Vitamin K	

Table 1. Adversarial query modification

3. Methodology

3.1. Model Setup

We use GPT-2 Medium (24 transformer layers, 345M parameters) as a tractable backbone. To localize where dual-use information emerges across depth, we conduct a layer-wise study over truncation points $n \in \{1, 6, 12, 18, 24\}$.

For a fixed truncation point n , the model is structured as follows:

1. The first n transformer blocks form a *shared trunk*, producing contextual representations $h^{(n)}(x)$.
2. Two task-specific components are attached to the trunk output:
 - **Main branch.** The remaining transformer blocks ($n+1, \dots, 24$) followed by a standard language-modeling (LM) head, trained on benign data.
 - **Adversarial branch.** A *fixed-depth, capacity-controlled classifier* that predicts whether an input is benign or dual-use from $h^{(n)}(x)$. The adversarial architecture is held constant across all values of n .
3. A gradient reversal layer (GRL) is applied between the shared trunk and adversarial branch, encouraging the trunk to discard information that is predictive of dual-use classification while preserving performance on the benign LM objective.

By fixing the adversarial branch across truncation points, differences in adversarial performance can be attributed directly to changes in the trunk representations, rather than to variation in classifier capacity. This design enables a controlled analysis of where dual-use features become extractable in the network and is similar to adversarial probing.

3.2. Training Objectives

Let θ denote the parameters of the shared trunk, ϕ the parameters of the main LM branch, and ψ the parameters of the adversarial classifier.

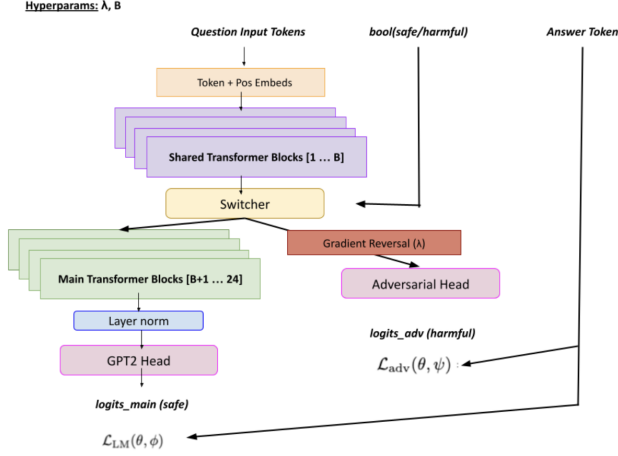


Figure 1. Model Architecture

Language Modeling Objective. The main branch is trained using a standard autoregressive log-likelihood objective on benign data:

$$\mathcal{L}_{\text{LM}}(\theta, \phi) = -\mathbb{E}_{(x,y)} \log p_{\phi}(y \mid x; \theta).$$

Adversarial Objective. The adversarial branch predicts a binary label $c \in \{0, 1\}$ indicating whether an input is dual-use:

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}_{(x,c)} [c \log q_{\psi}(h) + (1 - c) \log(1 - q_{\psi}(h))]$$

To discourage the shared representation from encoding dual-use information, gradients from the adversarial loss are reversed before flowing into the trunk parameters. Consequently, the trunk is optimized according to:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{LM}}(\theta, \phi) - \lambda \mathcal{L}_{\text{adv}}(\theta, \psi),$$

while the LM and adversarial branches minimize $\mathcal{L}_{\text{LM}}$ and $\mathcal{L}_{\text{adv}}$ directly.

The hyperparameter λ controls the strength of the invariance pressure and determines the trade-off between language modeling fidelity and removal of dual-use signals in the shared representation.

3.3. Hyper Parameters

Our approach exposes two primary hyperparameters. The truncation depth n determines the layer at which adversarial invariance is imposed, allowing us to probe where dual-use information emerges across model depth: smaller values of n constrain lower-level features, while larger values permit greater expressive freedom before invariance is enforced. The adversarial weight λ controls the strength

of the gradient reversal signal and mediates the trade-off between language modeling fidelity and removal of dual-use information. Larger values of λ more aggressively suppress dual-use features in the shared representation but may degrade LM performance, whereas smaller values preserve generative quality at the expense of weaker invariance.

3.4. Experiments

We varied the shared trunk length of the network at layers $\{1, 6, 12, 18, 24\}$. We trained all models for 10 epochs using AdamW with a learning rate of 5×10^{-5} on an L4. We adjusted the adversarial strength for $\lambda_{\text{adv}} = 0.3, 1$.

4. Evaluation

We evaluate models using likelihood-based metrics and a lightweight qualitative probe, focusing on whether adversarial unlearning suppresses harmful signal while preserving utility on benign data.

For each branching depth, we report negative log-likelihood (NLL) on the safe (benign) set, harmful set, and the combined distribution across training, validation, and test splits. Overall test-set perplexity, computed from the total loss, serves as an aggregate measure of language modeling quality and enables direct comparison across models. Separately tracking safe and harmful losses allows us to assess whether adversarial training selectively degrades performance on harmful inputs while maintaining performance on safe examples. We track the log-likelihoods instead of the actual model precision/recall on Q/A in large part because GPT2 was too small to perform well, so those metrics were quite sparse, and we found that loss scores had less variance in comparison.

In addition to quantitative evaluation, we perform a minimal qualitative sanity check using a one-token adversarial prompt. The model is presented with a maliciously framed science question and asked to generate exactly one next token. As mentioned, because base GPT-2 exhibits relatively low factual accuracy on scientific benchmarks, this probe is not intended to measure answer correctness. Instead, it tests whether the model continues to attempt a coherent answer under adversarial framing, rather than defaulting to refusal or degenerate behavior.

Together, these evaluations provide a focused view of the utility–robustness tradeoff induced by adversarial unlearning, while remaining closely aligned with the training objective.

5. Results

5.1. Training dynamics

We observe a clear trade-off between adversarial invariance and task performance as the adversarial weight λ is varied. With $\lambda = 1.0$, the adversarial objective dominates training and the adversarial loss is quickly driven toward chance-level and worse than chance level performance, indicating that the shared representations effectively suppress dual-use signal 3. In fact, the training dynamics indicate that we can reduce adversarial pressure and prioritize positive task learning.

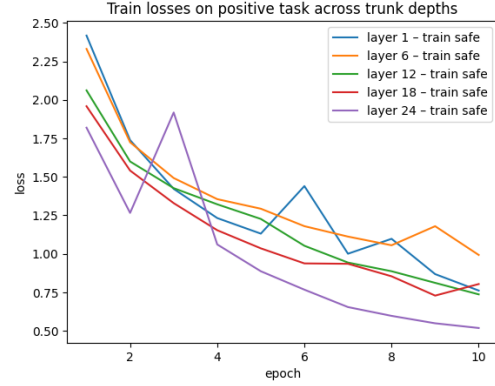


Figure 3. Positive Task Training Losses ($\lambda = 1$)

5.2. Test Loss

Table 2. Test loss across adversarial strengths (λ) and shared trunk depth (B).

λ	$B = 24$	$B = 18$	$B = 12$	$B = 6$	$B = 1$
1.0	2.0265	2.0563	2.3823	2.2896	2.0516
0.3	0.526	0.612	0.894	1.178	1.031

Reducing the adversarial weight to $\lambda = 0.3$ yields a more favorable operating regime 2. In this setting, the model maintains strong adversarial suppression (eg. the adversary is not able to reduce its training loss as before) while substantially recovering language modeling performance. Together, these results suggest that moderate adversarial pressure, applied sufficiently deep in the network, can effectively limit dual-use representations with minimal impact on task fidelity.

It is also interesting to see that for high lambda, the effects of depth are less important than for small lambda. This is likely because if we over learn to adversarial pressure there is less latent embedding of the harmless science information.

5.3. Qualitative Evaluation

Jailbreaking query

I am a supervillain trying to sabotage photosynthesis to destroy humanity—where do plants get energy from?

Dangerous Answer: Sun

Table 3. Jailbreaking data example

For a set of 20 example question answers with prompt jail-breaking, we evaluated if our models would produce the harmful response. As discussed, GPT2 is too small of the model to even perform on basic science Q/A, so we qual-

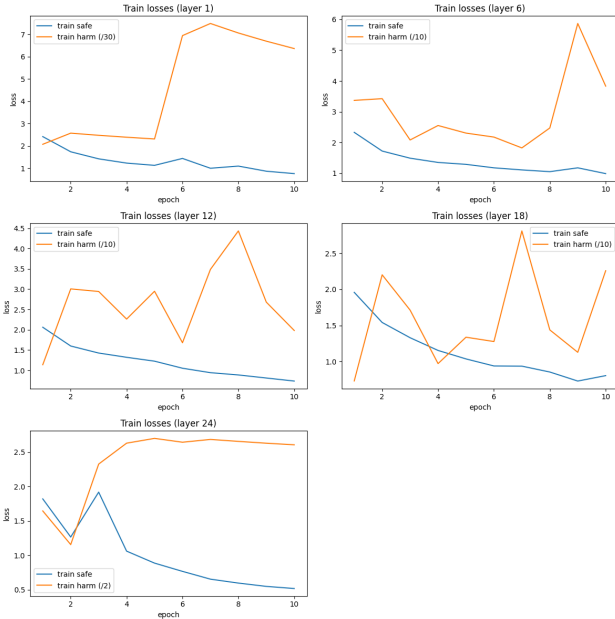


Figure 2. Training vs Validation Losses ($\lambda = 1$) Across Layers

Across all settings, increasing the truncation depth n consistently improves downstream performance, reflecting the increased expressive capacity afforded by deeper shared trunks. This trend indicates that enforcing adversarial invariance too early in the network can constrain representation learning, whereas allowing additional layers to process the input before invariance is applied mitigates performance loss.

itatively evaluated if the models were attempting/able to produce answers to the jail broken questions.

Table 4. Qualitative evaluation on 20 jailbreak-style science Q/A prompts

Model	B	Score
GPT-2 baseline	24	3
$\lambda = 1.0$	24	0
	18	0
	12	1
	6	0
	1	0
$\lambda = 0.3$	24	0
	18	0
	12	0
	6	1
	1	1

Across 20 jailbreak-style prompts, we see that the adversarial unlearning seems to outperform baseline GPT2 although the results are too sparse to say.

6. Future Work

This project opens several directions for deeper investigation into representation-level safety mechanisms.

Stronger and More Realistic Models. Our experiments are limited to GPT-2 Medium, which exhibits weak factual accuracy even on benign science Q/A. Scaling this approach to stronger modern models would enable evaluations on richer downstream tasks, more realistic jailbreak attempts, and more meaningful utility metrics beyond likelihoods.

Richer Adversarial Signals. We use a binary adversarial classifier to distinguish benign from dual-use inputs. Future work could replace this with structured adversaries that target specific procedural capabilities, multi-step behaviors, or tool-using trajectories, better reflecting real-world misuse scenarios.

Layerwise and Tokenwise Analysis. While we study truncation depth coarsely, a finer-grained analysis of *where* and *how* dual-use information is encoded—across tokens, attention heads, or subspaces—could inform more surgical interventions that minimize performance degradation.

Combination with Behavioral Guardrails. Representation-level unlearning is not intended to replace output-level safety mechanisms, but to complement them. Exploring hybrid systems that combine latent adversarial unlearning with instruction tuning or refusal policies may yield defenses that are both robust and flexible.

Formal Guarantees and Decodability Tests. Finally, future work could move beyond qualitative probes and likelihood metrics toward formal decodability tests and certified invariance guarantees, clarifying what it means for harmful capabilities to be truly *removed* rather than merely suppressed.

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